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Deep learning-assisted screening and diagnosis of scoliosis: segmentation of bare-back images via an attention-enhanced convolutional neural network

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Abstract

Background Traditional diagnostic tools for scoliosis screening necessitate a substantial number of specialized personnel and equipment, leading to inconvenience that can result in missed opportunities for early diagnosis and optimal treatment. We have developed a deep learning-based image segmentation model to enhance the efficiency of scoliosis screening.

Methods A total of 350 patients with scoliosis and 108 healthy subjects were included in this study. The dataset was created using their bare back images and standing full-length anteroposterior spinal X-rays. An attention mechanism was incorporated into the original U-Net architecture to build a Dual AttentionUNet model for image segmentation. The entire dataset was divided into the training (321 cases), validation (46 cases), and test (91 cases) sets in a 7:1:2 ratio. The training set was used to train the Dual AttentionUNet model, and the validation set was used to fine-tune hyperparameters and prevent overfitting during training. The performance of the model was evaluated in the test set. After automatic segmentation of the back contour, a back asymmetry index was calculated via computer vision algorithms to classify scoliosis into different severities. The accuracy of classifications was statistically compared to those of three clinical experts.

Results Following the segmentation of bare back images and the application of computer vision algorithms, the Dual AttentionUNet model achieved an accuracy, precision, and recall rate of over 90% in predicting severe scoliosis. Notably, the model achieved an AUC value of 0.93 in identifying whether the subjects had scoliosis, which was higher than the 0.92 achieved by the deputy chief physician. In identifying severe scoliosis, their AUC values were 0.95 and 0.96, respectively.

Conclusion The Dual AttentionUNet model, based on only bare back images, achieved accuracy and precision comparable to clinical physicians in determining scoliosis severity. Radiation-free, cost-saving, easy-to-operate and noninvasive, this model provides a novel option for large-scale scoliosis screening.

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Keywords Scoliosis, Artificial intelligence, Deep learning, Image segmentation, Screening

Introduction

Scoliosis, typically referred to the lateral curvature of the spine [1], is diagnosed when the Cobb angle reaches or exceeds 10° on standing X-rays. Classified according to the Cobb angle, mild scoliosis requires regular monitoring and exercise, moderate scoliosis may necessitate bracing along with physiotherapy, and severe scoliosis often requires surgical intervention [2]. Due to its subtle symptoms in the early stage, scoliosis diagnosis is often missed and treatment delayed, potentially leading to adverse events [3, 4].

Traditional screening of scoliosis relies on comprehensive physical examinations and radiographic measurements, which are time-consuming and labor-intensive [5]. Furthermore, subjective errors may interfere with the diagnosis and treatment process. Full-length spinal X-rays are accurate, but pose radiation-induced injury, particularly for adolescents. In addition, scoliosis may aggravate if treatment delays, leading to serious complications [6, 7]. Therefore, safe and efficient tools remain to be invented for early scoliosis screening and intervention.

Machine learning (ML) enables the automatic learning and extraction of useful information from large datasets, which can be used in clinical decision-making [8, 9]. Deep learning (DL), a subfield of ML, excels in handling large unstructured data, such as images [10, 11]. Based on artificial neural networks (ANNs), DL incorporates prior constraints or regularization to prevent overfitting and enhance generalization. Network weights are automatically learned from training data, thus minimizing human subjectivity and improving diagnostic accuracy.

Convolutional neural networks (CNNs) function as the core part in DL processes [12]. Consisting of convolutional layers, pooling layers, and fully connected layers, CNNs have demonstrated outstanding performance in identifying and segmenting regions of interest on images. Besides, utilizing fully convolutional layers and transposed convolution layers, CNNs can also upsample feature maps to ensure spatial consistency between outputs and input images, thereby achieving precise pixel-level segmentation. The U-Net architecture relies on downsampling to extract global features and upsampling to refine feature maps, thus enabling accurate pixel-level segmentation [13]. Attention U-Net, a version of U-Net enhanced by incorporating attention mechanisms, can further improve segmentation accuracy, especially on complex medical images.

In recent years, AI technology has revolutionized the diagnosis and treatment of orthopedic diseases [14], such as spinal deformities [15]. In this study, we established a Dual AttentionUNet model using only bare back

photographs, to perform preliminary screening and classification for scoliosis.

Materials and methods

Data collection and grouping

A total of 458 participants were enrolled in this study, including 350 with scoliosis who were treated in outpatient and inpatient departments from April 2019 to June 2023, as well as 108 healthy individuals. The participants were recruited from Ningxia Medical University General Hospital, Henan Provincial People's Hospital, and Zhengzhou First People's Hospital. This study employed a dataset that was relatively small for AI analysis; nevertheless, even smaller medical datasets have been known to yield successful experimental outcomes. For instance, Zafer Soydan et al. [16] have collected the real-world data of 81 patients' LCPD (Leg-Calve-Perthes Disease) hip X-rays, and constructed a CNN classification model, which achieved an accuracy of 83.33% in the verification set; in their study, the model's performance was compared with those of 11 doctors, showing that only two doctors achieved a higher classification accuracy (both at 84%) than the CNN model. Certainly, in the future, with the construction of larger and more comprehensive datasets, the general applicability of our DL model can be further proved, and its bias reduced.

Inclusion criteria included (1) no history of spinal surgery and (2) complete full-length spinal X-rays and bare back images. Exclusion criteria included (1) pseudoscoliosis caused by pain, poor posture, or pelvic imbalance and (2) incomplete or low-quality bare back images or full-length spinal X-rays.

Standard Cobb angle measurements were obtained from the full-length spinal X-rays by two spine surgeons with ten years of surgical experience. Any disagreements were resolved through discussion with a senior spine surgeon with over 25 years of experience in scoliosis treatment, until a consensus was reached among the three experts. Cobb angles were divided into four subsets: healthy, Cobb angle $<10^\circ$ (108 cases); mild, $10^\circ \leq$ Cobb angle $<20^\circ$ (113 cases); moderate, $20^\circ \leq$ Cobb angle $<45^\circ$ (120 cases); severe, Cobb angle $\geq 45^\circ$ (117 cases). For the effective training, optimization, and evaluation of the Dual AttentionUNet model, the subjects were randomly divided into training (321 cases), validation (46 cases), and test (91 cases) sets in a ratio of 7:1:2 (Table 1). To further investigate the similarity and normality of these three sets, we plotted normal probability plots (Q-Q plots) to visualize the distributions of characteristics (Fig. 1).

Table 1 Distributions of basic characteristics in the training, validation, and test sets

Item	Training Set (n = 321)	Validation Set (n = 46)	Test Set (n = 91)	Chi-Square	PValue
Gender (Male/Female)	107/214	14/32	29/62	0.194	0.908
Age (Years)	13.66 ± 4.567	13.17 ± 4.725	13.64 ± 4.900	\	0.799
Severity (Healthy/Mild/Moderate/Severe)	74/87/81/79	11/10/12/13	23/16/27/25	3.842	0.698
Disease Status (Healthy/Scoliotic)	74/247	11/35	23/68	0.197	0.906

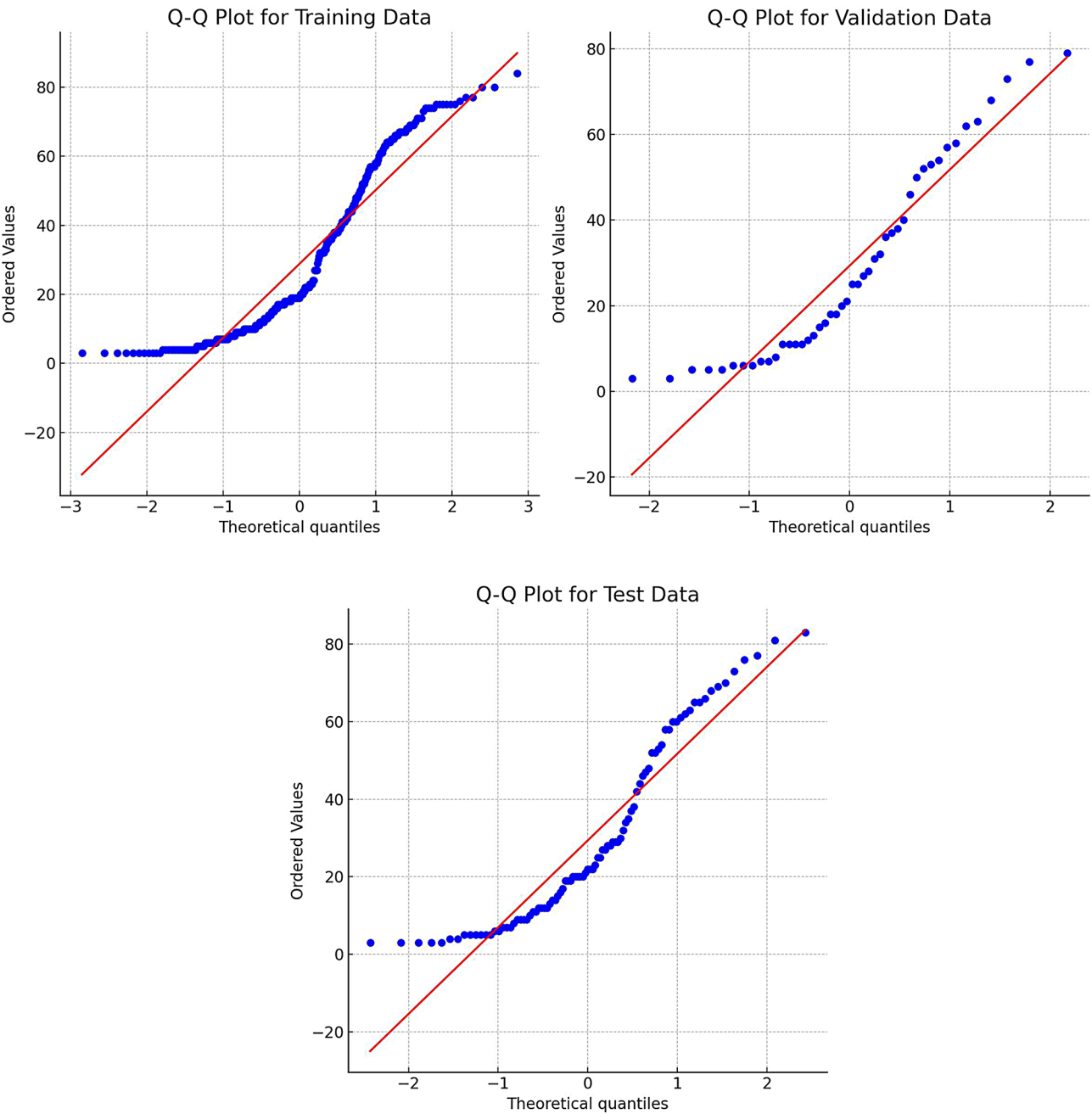


Fig. 1 Normal probability plot (Q–Q plot) of the true Cobb angles in the training, validation, and test sets

To compare the performance of the our DL model with that of clinicians, we invited three spine surgeons from our department with varying years of clinical experience (a deputy chief physician, an attending physician, and a resident physician) to conduct a single-blind evaluation of 458 bare back images. They judged the severity of scoliosis subjectively in all patients through visual inspection within 40 min (5.24 s per image, which was significantly

longer than the 1.51 s per image achieved by the Dual AttentionUNet model combined with computer vision algorithms).

This study was conducted in accordance with the Declaration of Helsinki and received approval from the hospital's ethics committee (the Medical Science Research Ethics Review Committee of Ningxia Medical University General Hospital, KYLL20220133). All participants provided informed consent and signed consent forms before their bare back images were collected, ensuring the legality and ethical integrity of the study.

X-ray examination and back imaging

Standing full-length anteroposterior spinal X-rays were obtained by professional radiologists via a Philips Ingenuity Care X-ray system, with participants standing naturally, arms at their sides.

For bare back imaging, participants were instructed to expose their back to the level of the posterior superior iliac spine, with their feet together and arms either at their sides or crossed in front of the chest. A camera (Canon EOS R50) with an RF-S 18–45 mm f/4.5–6.3 IS STM lens was used, featuring an effective pixel count of 24.2 million. The camera was positioned 2 m from the subject, and its position and focus were adjusted to ensure the acquisition of high-resolution images of the exposed back.

Enhancement of the U-Net model

Figure 2 presents the updated structure of our developed Dual AttentionUNet, with Fig. 2b illustrating the comparison between the original and improved network architectures. Specifically, the original U-Net used only 3×3 convolutions for feature extraction, lacking an explicit mechanism for feature selection. This resulted in its inability to dynamically adjust based on the contribution of different channels, leading to potential oversight of important features or undue focus on redundant ones.

To address this, we enhanced the U-Net structure by incorporating channel attention and self-attention mechanisms to dynamically adjust feature importance. The channel attention mechanism allowed for dynamic weight adjustment based on the importance of features in each channel, enabling the network to automatically select and focus on the most valuable features for the task. Meanwhile, the self-attention mechanism modeled long-range dependencies among input features, enhancing the network's capability to process global information.

Processing pipeline

In this study, ground truth (GT) labels were manually annotated by two spine surgeons, both with eight years of clinical experience, to provide data for supervised learning. The Dual AttentionUNet model was trained to

segment bare-back images, locate key regions, and delineate the back contour. The segmentation data served as the basis for scoliosis detection. After segmentation, the images were normalized, and different pixel values were filled inside the back contour to highlight the regions of interest (ROIs). Computer vision algorithms were used to draw parallel lines on both sides of the neck, dividing the contour into left and right parts for comparison, with an asymmetry index calculated with the following formula:

$$S = \frac{(Left - Right)}{(Left + Right)} \quad (1)$$

where the value of S ranges from -1 to 1 . Values closer to 0 indicate good symmetry, while values approaching 1 or -1 indicate poor symmetry.

Statistical methods

Various statistical methods in Python 3.12.6 were employed to evaluate the performance of the Dual AttentionUNet model in classifying scoliosis. First, a Spearman correlation analysis was performed to examine the relationship between the back asymmetry index and scoliosis severity [17]. The Spearman correlation coefficient, typically denoted by the Greek letter ρ (rho), was used to quantify the monotonic relationship between two variables. The correlation coefficient ranged from -1 to $+1$, where $+1$ representing a perfect positive correlation, -1 represents perfect negative correlation, and 0 indicates no correlation.

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} \quad (2)$$

where n is the number of data points, and d is the rank difference for each data point.

In addition, a confusion matrix was used to analyze the accuracy, precision, and recall of the Dual AttentionUNet model [18]. The closer the accuracy, precision, and recall values are to 100% , the more accurate the method is in approximating the true results. The formulas for these metrics are as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \quad (3)$$

$$Precision = \frac{TP}{TP + FP} \times 100\% \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \times 100\% \quad (5)$$

where:

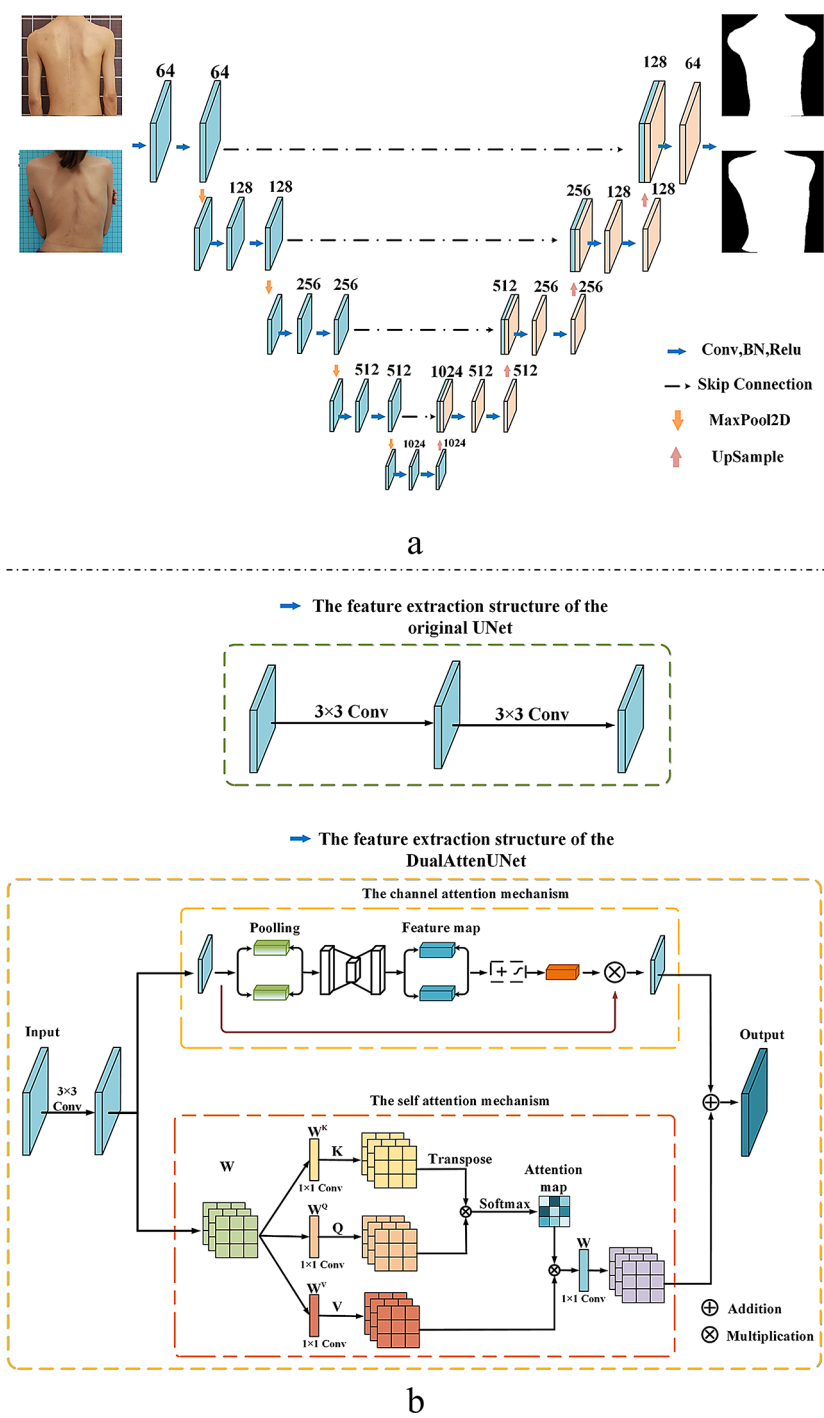


Fig. 2 Dual AttentionUNet. **(a)** Network Architecture Diagram. **(b)** Detailed Schematic of U-Net Improvements

TP (true positive) represents correctly predicted positive cases,

TN (true negative) represents correctly predicted negative cases,

FP (false positive) represents incorrectly predicted positive cases,

FN (false negative) represents incorrectly predicted negative cases.

To compare the model's discriminative ability with that of clinicians, Cohen's kappa value (κ) was calculated to describe the agreement between the predicted values and actual values [19]:

Kappa = 1: perfect agreement.

Kappa = 0: the agreement is equivalent to what would be expected by chance.

$$\kappa = \frac{p_0 - p_e}{1 - p_e} \quad (6)$$

where:

where p_0 represents the observed agreement (the proportion of instances where the raters provided the same rating).

p_e is the expected agreement (the proportion of agreement expected by random chance).

Finally, the area under the receiver operating characteristic curve (AUC) was used to compare the model's performance with that of clinicians. A higher AUC value (closer to 1) indicates better performance.

Experiments and results

Experimental environment

The Dual AttentionUNet model was implemented using Python 3.12.6 and PyTorch 2.4.1 and trained on an NVIDIA RTX 4060 GPU. The Adam optimizer was used with an initial learning rate of 0.001 and a weight decay of $1e-4$. A batch size of 8 was used for all models, and the training was conducted for 100 epochs.

Image segmentation and computer vision algorithm processing

Since most DL models require square input images, the original images (Fig. 5a) were preprocessed by padding, cropping, and denoising (Fig. 5b). The Dual AttentionUNet model was then trained on the training set and compared with the traditional U-Net network (Figs. 3

and 4). The parameters that achieved the best performance were saved as the optimal network weights. In this study, ten training runs were driven for each model (Dual AttentionUNet and U-Net), saving the optimal network parameters from each run. The averages of the optimal network parameters from the ten runs were calculated to produce the statistical data presented in Table 2. The validation set was used to adjust hyperparameters and prevent overfitting during training. The test set was used to evaluate the model's performance, and contour images were automatically segmented (Fig. 5c). The segmentation results of Dual AttentionUNet closely resembled the manual labels by physicians, as compared to traditional UNet segmentation (Fig. 6). The figure illustrated that the Dual AttentionUNet was more sensitive to boundary information, producing clearer and more continuous segmentation boundaries, closer to the true label image.

Spearman correlation analysis

The results of the correlation analysis are presented in Table 3, and the scatter plot with the fitted line is shown in Fig. 7.

The Spearman correlation coefficient, calculated using Python 3.12.6, was 0.806 ($P < 0.01$), indicating a significant positive correlation between the back asymmetry index and scoliosis severity. Subsequently, the reference range of the dorsal asymmetry index for each severity category was determined using interval estimation methods. Because of this positive correlation, the severity could be estimated by calculating the dorsal asymmetry index.

Discriminative performance of the dual AttentionUNet model

The confusion matrix, generated by comparing the ground truth labels with the predicted labels using Python 3.12.6, is shown in Fig. 8. This figure used '0, 1, 2, 3' to represent 'healthy spine, mild scoliosis, moderate scoliosis, and severe scoliosis', respectively. The numbers in Fig. 8a represented sample counts, and the percentages in Fig. 8b indicated their proportions. A darker color corresponded to a larger sample count or proportion. Overall, the model achieved an accuracy of 86.9%, a precision of 87.2%, and a recall of 86.9%. Observing this figure, it can be noted that patients with mild scoliosis had higher rates of false positives and false negatives, suggesting



Fig. 3 Feature maps visualized during the training of the Dual AttentionUNet model

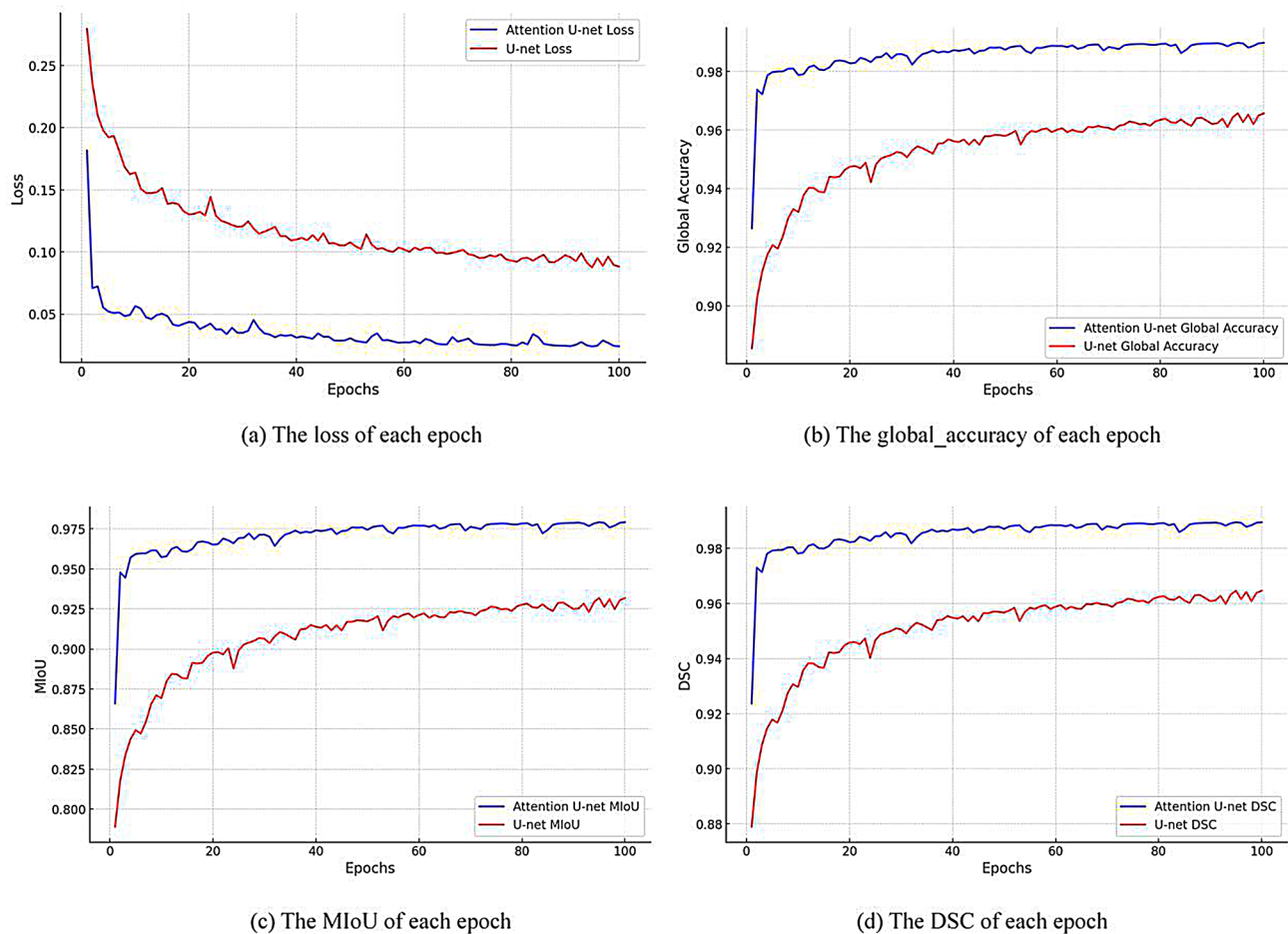


Fig. 4 Training and performance of the Dual AttentionUNet model, compared with the traditional U-Net

Table 2 Quantitative comparison between the training of different networks

Method	MIoU/%	DSC/%	Global accuracy/%
Dual AttentionUNet	97.91 ± 1.125	98.95 ± 1.009	98.98 ± 0.065
U-Net	93.19 ± 1.376	96.46 ± 1.237	95.76 ± 0.122

a need for more robust feature extraction methods for patients with less pronounced spinal deformities to improve diagnostic accuracy.

The classification results of the Dual AttentionUNet model are summarized in Table 4; Fig. 9, with the model achieving an accuracy exceeding 90%.

Performance comparison between the model and clinicians

Using Python 3.12.6, Cohen's kappa value was calculated to describe the agreement between the predicted values and the ground truth. The Dual AttentionUNet model achieved a Cohen's kappa value of 0.83, indicating that its results closely reflected the true conditions (Fig. 10).

ROC curves were plotted via Python 3.12.6 to illustrate the ability of the Dual AttentionUNet model and clinicians to distinguish scoliosis based on bare-back images (Fig. 11). The predictive ability of the Dual AttentionUNet model across the four severity levels was

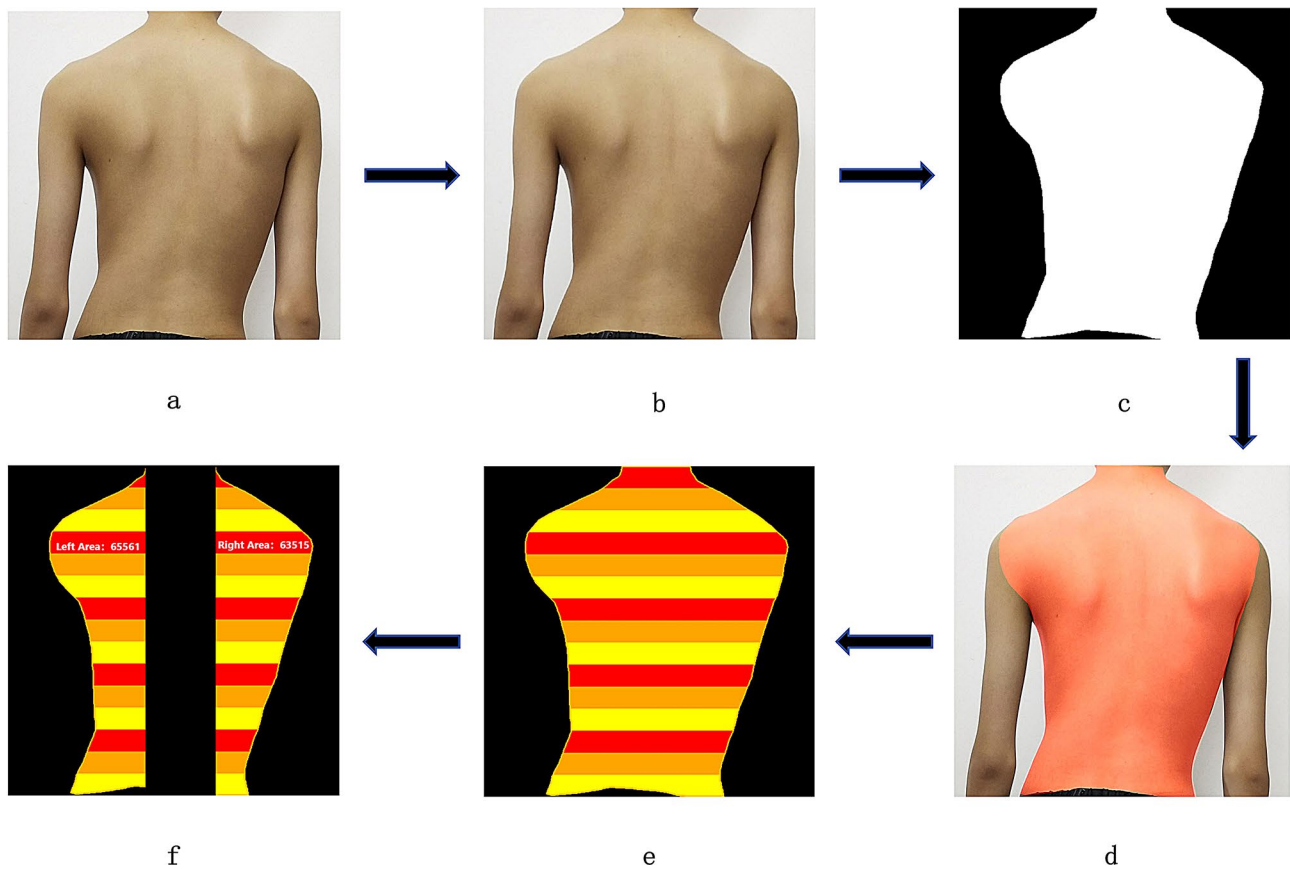


Fig. 5 Schematic diagram of image segmentation by the Dual AttentionUNet model and computer vision algorithms

comparable to that of the deputy chief physician. The Dual AttentionUNet model achieved AUC values of 0.93 for distinguishing healthy individuals and 0.95 for identifying severe cases (Table 5). Given that the ability to predict and distinguish healthy individuals from scoliotic patients is essentially the screening capability in clinical practice, it could be concluded from Fig. 11.a that the screening capability of the Dual AttentionUNet model was almost equivalent to that of the deputy chief physician.

Discussion

This study investigated the application of attention-enhanced convolutional neural network (CNN) models, specifically Dual AttentionUNet, in scoliosis screening

and severity assessment based on bare-back images. This approach provides a radiation-free alternative to traditional physical examination and X-ray imaging while maintaining high diagnostic accuracy and efficiency [20–24]. By integrating attention mechanisms into the U-Net architecture, the model achieves notable segmentation accuracy, enabling precise detection of back asymmetry and scoliosis severity. Remarkably, the Dual AttentionUNet demonstrated performance comparable to experienced clinicians in classification tasks, as evidenced by a high AUC (0.927) and Cohen's kappa score (0.83). These findings suggest that the model can reliably assist in clinical decision-making, particularly in resource-limited settings or large-scale screening programs.

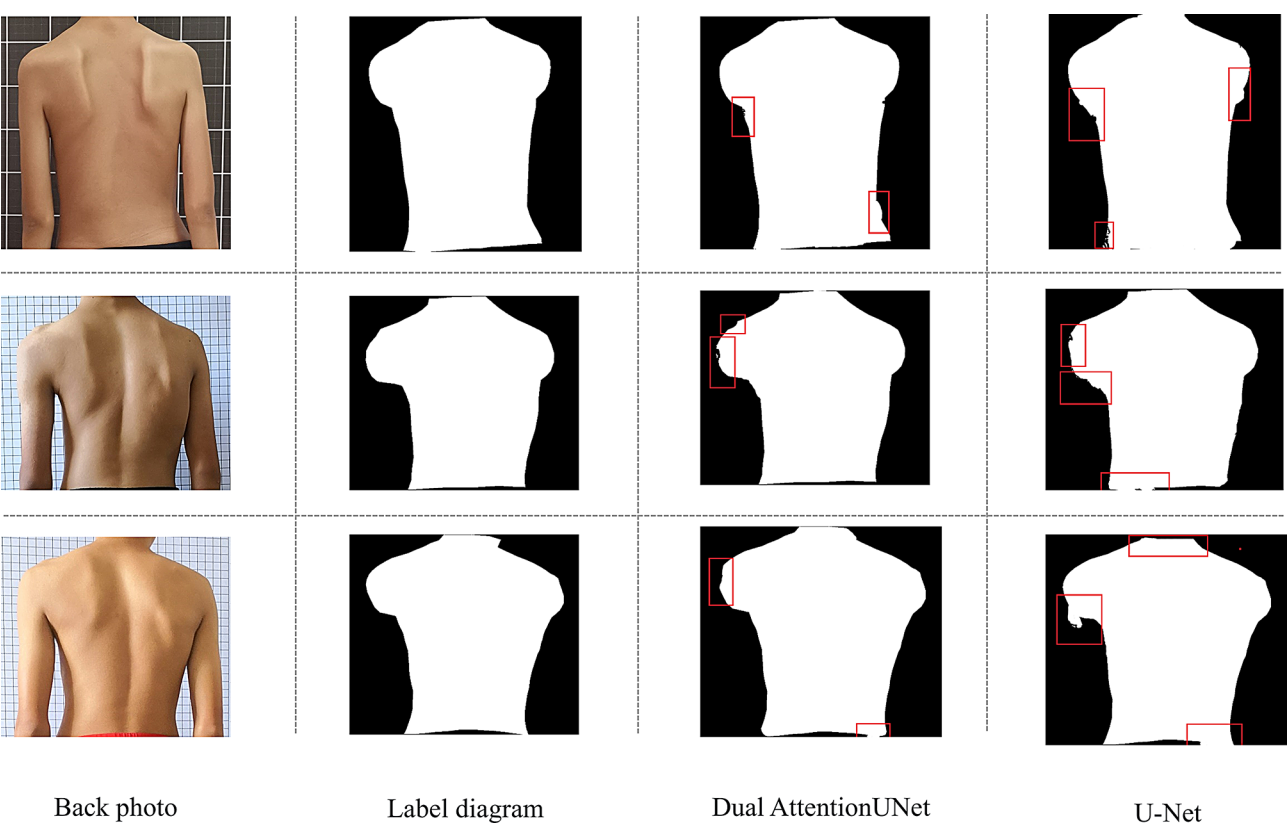


Fig. 6 Visualization of segmentation results, from left to right: back image, label image, Dual AttentionUNet segmentation, and U-Net segmentation

Table 3 Correlations between the asymmetry index (absolute value) of the back contour and the severity of scoliosis

	Asymmetry index (absolute value)	Severity of sco- liosis
Asymmetry index (absolute value)	1	0.806**
Severity of scoliosis	0.806**	1

Note: ** indicates a significant correlation at 0.01 (two-tailed)

The high performance of the Dual AttentionUNet model in scoliosis diagnosis, comparable to that of a deputy chief physician, is attributed to its advanced design, which integrates attention mechanisms into the traditional U-Net architecture. These mechanisms enable the model to dynamically focus on key regions within the image, enhancing segmentation accuracy and reducing noise from irrelevant areas. By leveraging global and local feature representations, the model achieves precise pixel-level segmentation, crucial for identifying subtle back asymmetries indicative of scoliosis. Furthermore, the training process optimizes the model's parameters through a balanced dataset and rigorous hyperparameter tuning, ensuring its capability of robust generalization. The combination of this innovative architecture and

systematic optimization enables the model to reach diagnostic accuracy and consistency that often surpass those of human experts. Compared to human experts, the model exhibits significant advantages in efficiency, consistency, and scalability. While human assessment is susceptible to fatigue, variability, and subjective judgment, the model operates consistently, unaffected by external factors. It enables rapid and accurate evaluations within seconds, substantially reducing the time required for large-scale screenings. Additionally, the model reduces variability among clinicians of different experience levels, minimizing the risk of misdiagnosis or inconsistent assessments. Its seamless integration into mobile platforms provides a wide accessibility, making it cost-effective and reliable for scoliosis diagnosis across diverse populations and settings. These attributes position the model as a transformative innovation for improving early detection and management of scoliosis.

The innovation of this study lies in the introduction of an attention mechanism, enabling the Dual AttentionUNet to focus on key features of medical images, thereby outperforming the traditional U-Net model in terms of precision and localization. Deep learning (DL) techniques have been extensively applied in the screening

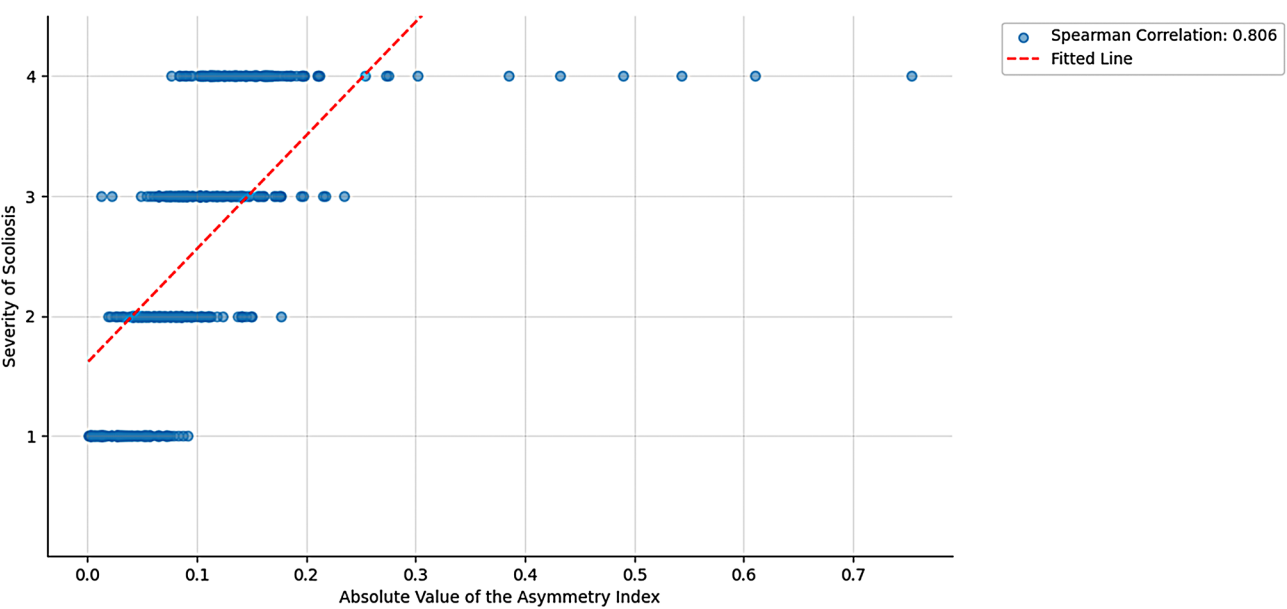


Fig. 7 Scatter plot and fitting line of the Spearman correlation analysis

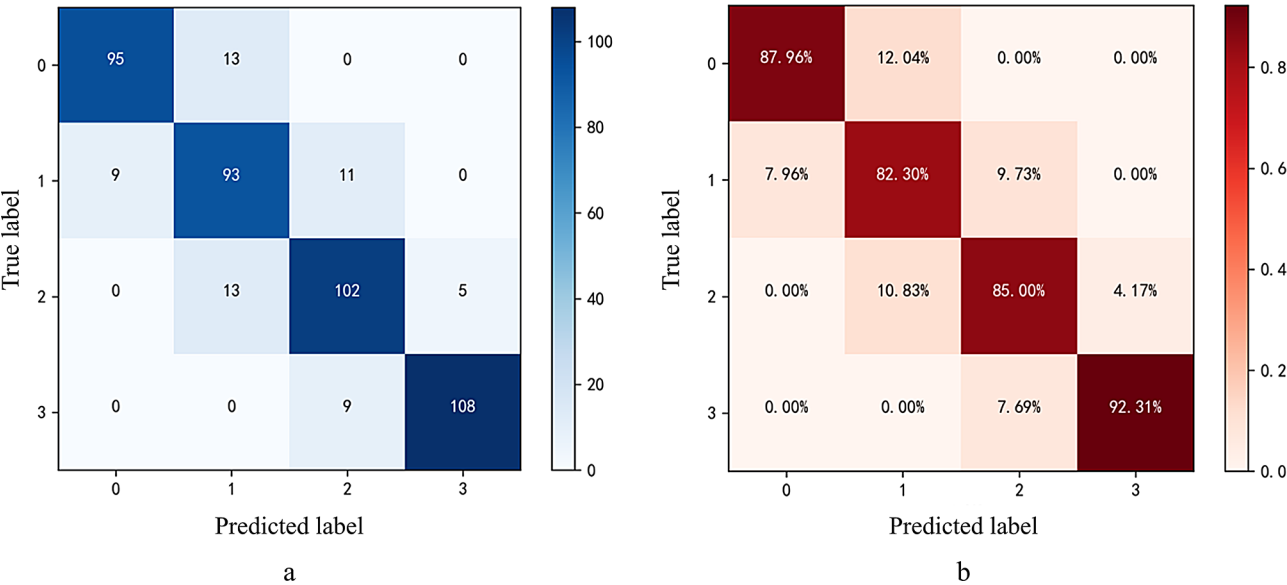


Fig. 8 Confusion matrix generated by true labels and predicted labels. **(a)** Sample counts. **(b)** Sample count ratios

Table 4 Performance (%) of the dual AttentionUNet model ins classifying scoliosis

Severity	Accuracy (%)	Precision (%)	Recall (%)
Healthy	95.2	91.4	88
Mild	90	78.2	82.3
Moderate	91.7	83.6	85
Severe	96.9	95.6	92.3

and diagnosis of spinal deformities, particularly in the analysis of X-ray, CT, and MRI data [25–28]. Galbusera

et al. [29] proposed a fully automated DL method for spinal radiological analysis, achieving a high accuracy in the computation of clinical parameters, such as thoracic kyphosis and scoliosis angles. However, the large error range (2.7° to 11.5°) underscores the need for further optimization. Similarly, DL models like ResNet50 have shown promise in distinguishing benign from malignant spinal fractures on CT images, with sensitivity and accuracy reaching 90% and 85%, respectively, though validation with larger datasets is required [30]. Research

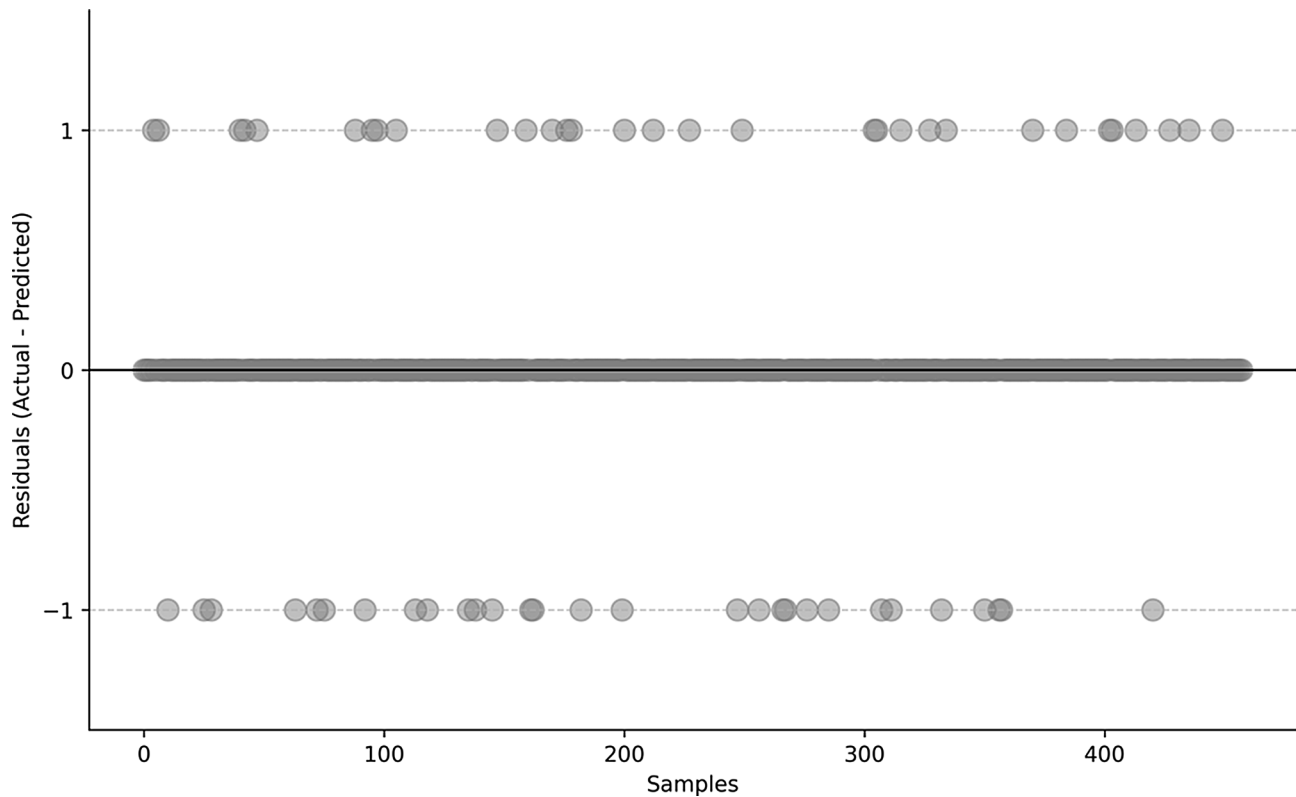


Fig. 9 Distribution residual plot of the Dual AttentionUNet model

on lumbar MRI has also indicated that the combination of DL and machine learning (ML) techniques can enhance classification performance, as demonstrated by DenseNet201 and SVM, which achieved higher precision and F1 scores [31].

Despite the promising potential of AI in scoliosis diagnosis, its limitations may affect the model's generalizability and overall performance. One key limitation is the relatively small dataset, which may not fully capture the variability of scoliosis cases across diverse populations. A larger, more comprehensive dataset would be essential for improving the model's ability to accurately identify different types and severities of scoliosis. Another limitation is the model's lower performance in patients who

have higher body mass indexes (BMI) or less pronounced deformities. In such cases, the asymmetry may be subtle under naked eyes, leading to a decrease in the accuracy of scoliosis detection. A more favorable model would incorporate additional factors, such as BMI, and more advanced algorithms, like multiscale neural networks [32, 33]. Lastly, equipping our model on mobile platforms requires significant technological support, which is not available in some areas. Therefore, while the study demonstrates the potential of AI in scoliosis diagnosis, overcoming these limitations through dataset expansion, algorithm optimization, and technological improvements is critical for its successful implementation and broader use.

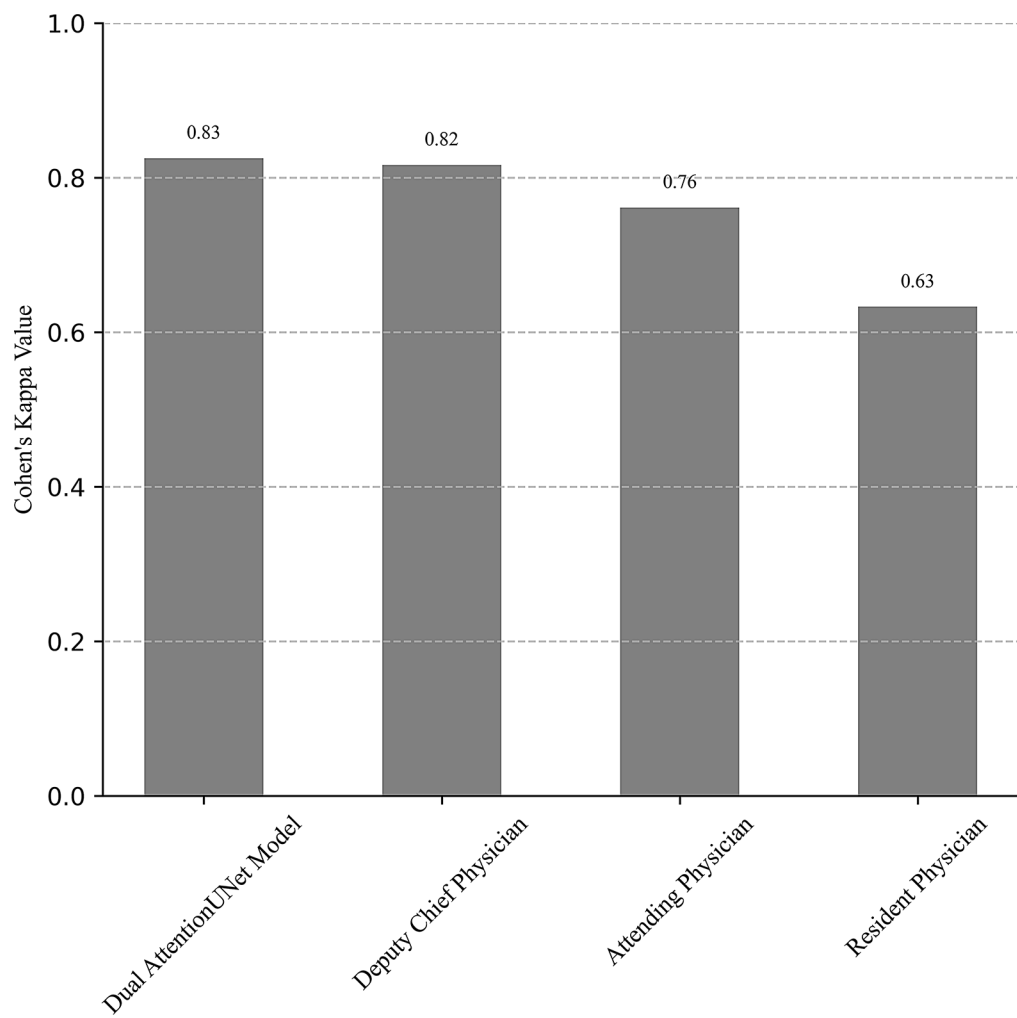


Fig. 10 Comparison of Cohen's kappa values between the Dual AttentionUNet model and clinicians with actual values

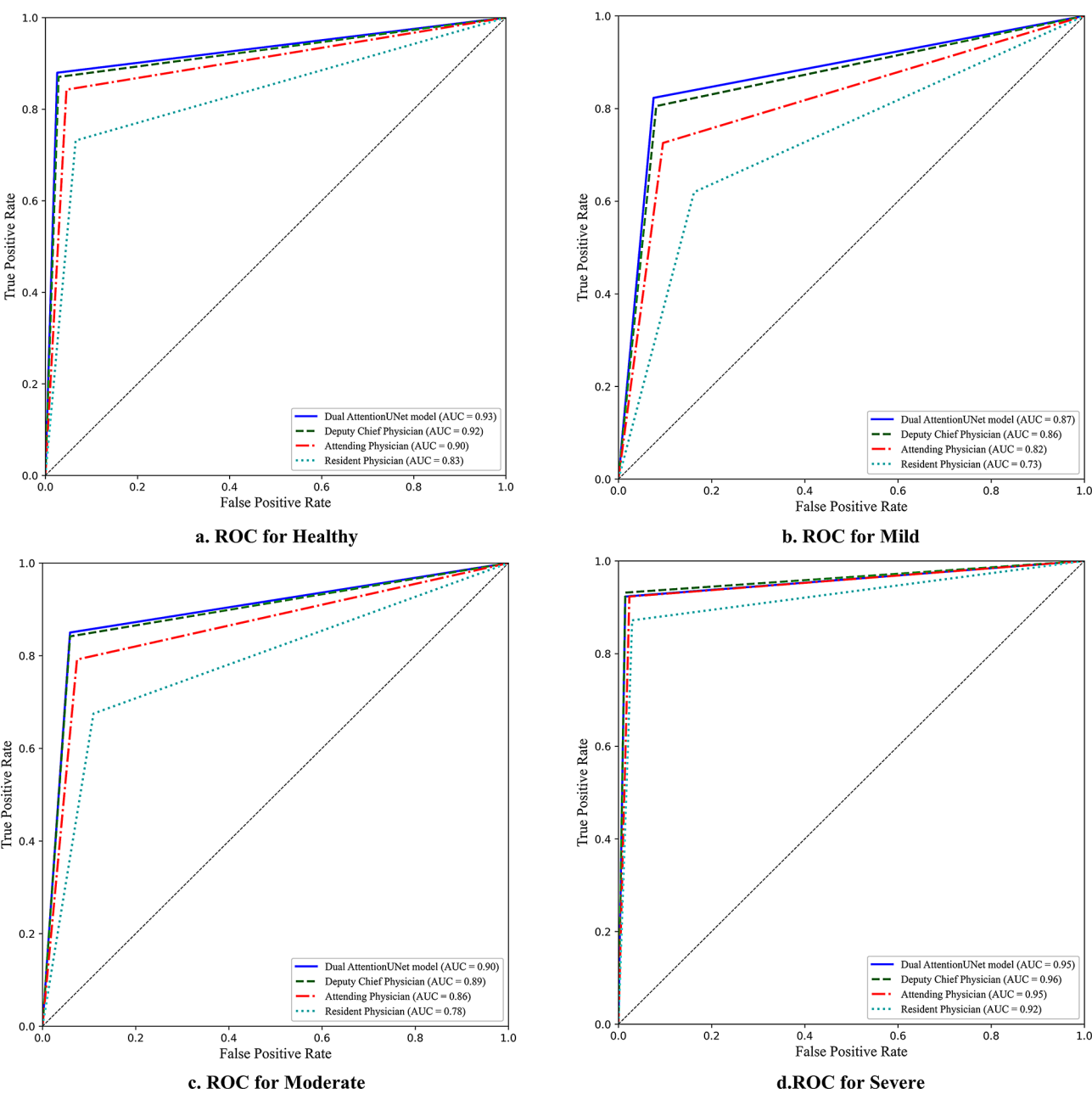


Fig. 11 ROC curves of the Dual AttentionUNet model and clinicians in distinguishing scoliosis of various severities

Table 5 AUC values of the dual AttentionUNet model and clinicians in distinguishing scoliosis of various severities

AUC	Dual AttentionUNet Model	Deputy Chief Physician	Attending Physician	Resident Physician
Healthy	0.927	0.921	0.898	0.832
Mild	0.874	0.862	0.815	0.729
Moderate	0.895	0.891	0.859	0.783
Severe	0.954	0.958	0.950	0.921

Conclusion

The Dual AttentionUNet model in this study offers an efficient, cost-effective, and easy-to-use method for scoliosis screening and diagnosis. As technology advances and more data is accumulated, its application in spinal health is expected to become more widespread, contributing significantly to public health initiatives.

Acknowledgements

Not applicable.

Author contributions

XYD and XJM were responsible for data collection and drafting the primary manuscript. MQZ and LNW contributed to the refinement, and conducted a review for grammatical and spelling errors. DQY and LLD participated in the collection of partial data. NKN provided guidance on the innovation of the writing, and ensured the scientific accuracy and rigor of the paper.

Funding

This project is financially supported by the National Natural Science Foundation of China (No. 82260436); Key R&D program of the autonomous region (No. 2022BEG03099); The Ningxia Provincial Key Laboratory for Clinical Pathogens and Microbiology Open Research Program (No. MKLG-2024-06).

Data availability

If you require the raw data, please contact the corresponding author at niuningkui@163.com.

Declarations

Ethics approval and consent to participate

The study was approved by the Ethics Review Committee of the General Hospital of Ningxia Medical University (No. KYLL20220133). All procedures were conducted in accordance with the ethical standards established by the Declaration of Helsinki and its subsequent amendments. Written informed consent was obtained from all participants.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Received: 9 November 2024 / Accepted: 3 February 2025

Published online: 14 February 2025

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